







Validating Causality





S



-

R  +



R



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S



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C



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R  +

R.B. -

C  +



C



—





po

po









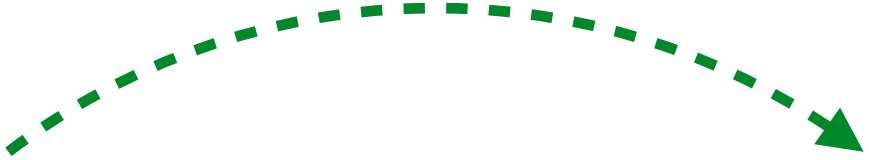
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po

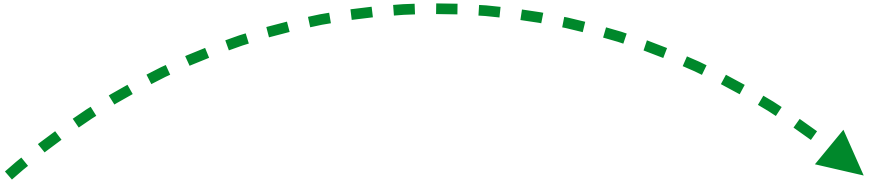


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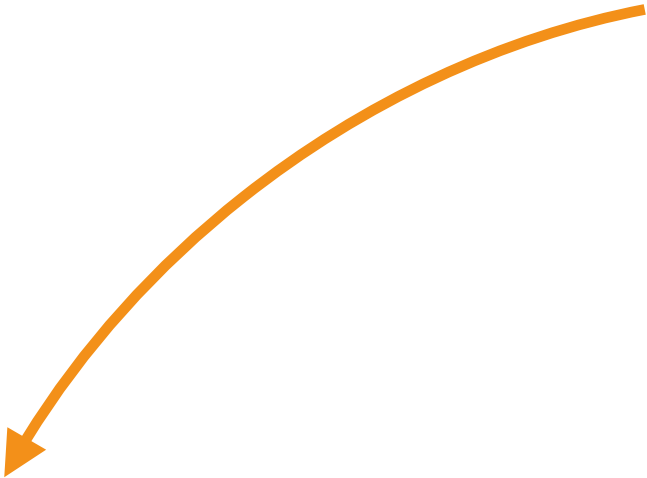
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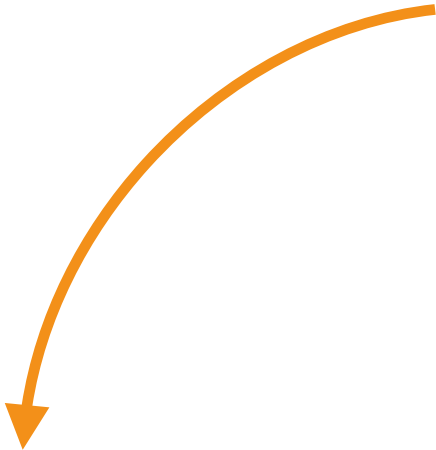




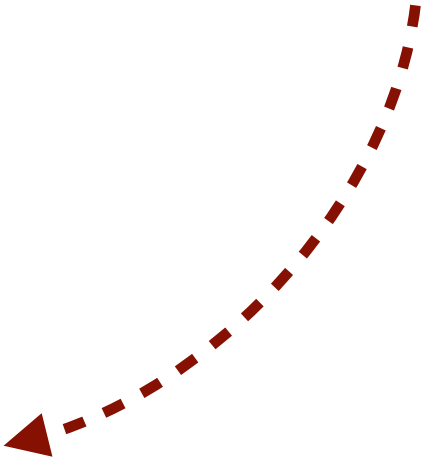


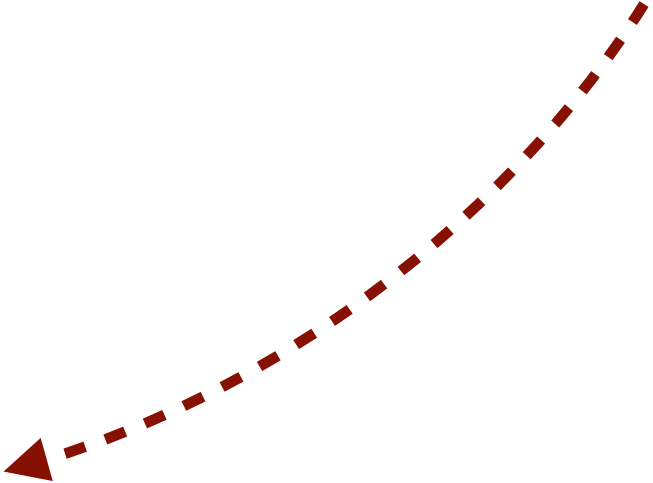


cosrc



collog





hb

h

b



```
update(src, +100)
```

```
update(src, -100)
```



```
def update(account: cown[Account], amount: int) {  
  when(account) { A  
    if(account.balance + amount > 0) {  
      account.balance += amount  
      val id = account.id()  
      when(log) { B log.record(id, amount) }  
    } } }  
}
```

R  +

R



—









po

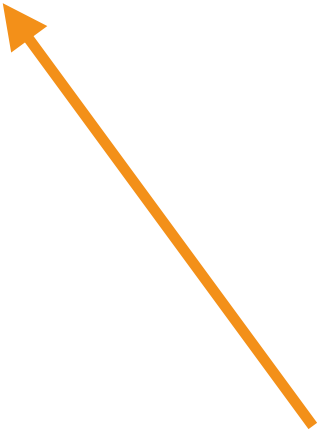
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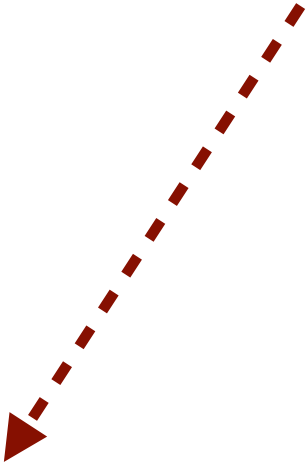
o



collog









h

b





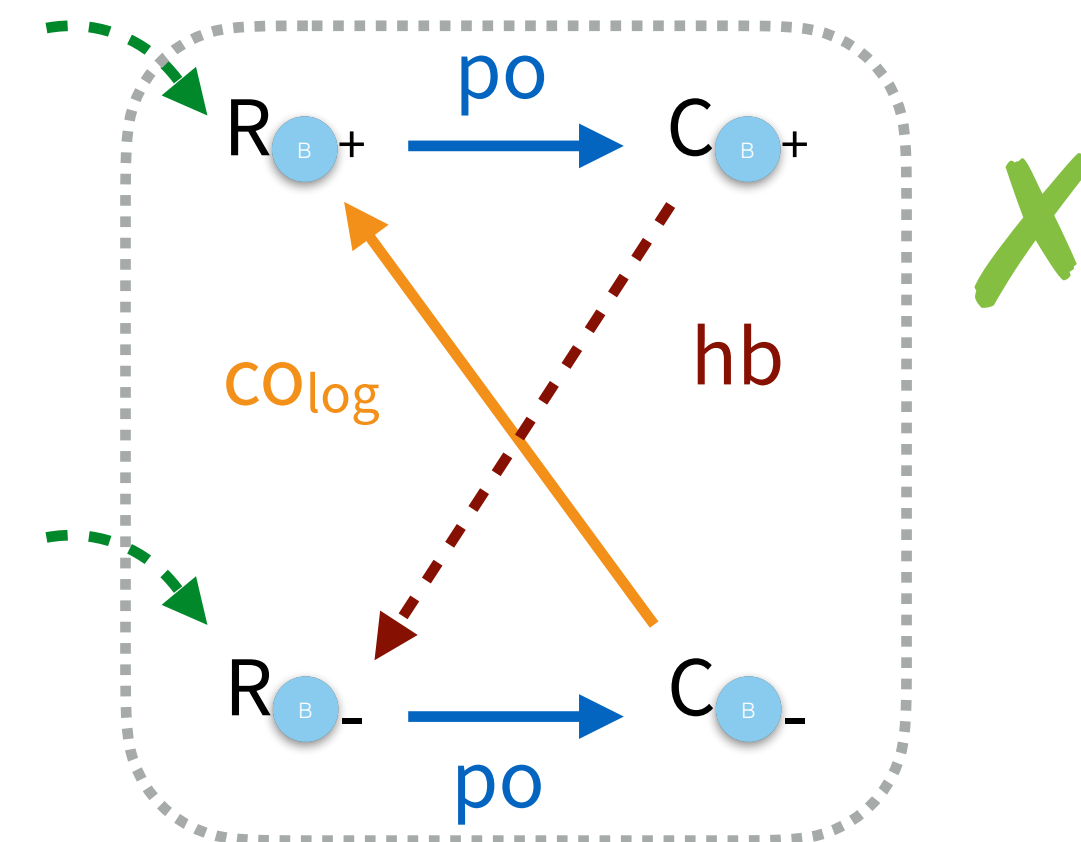
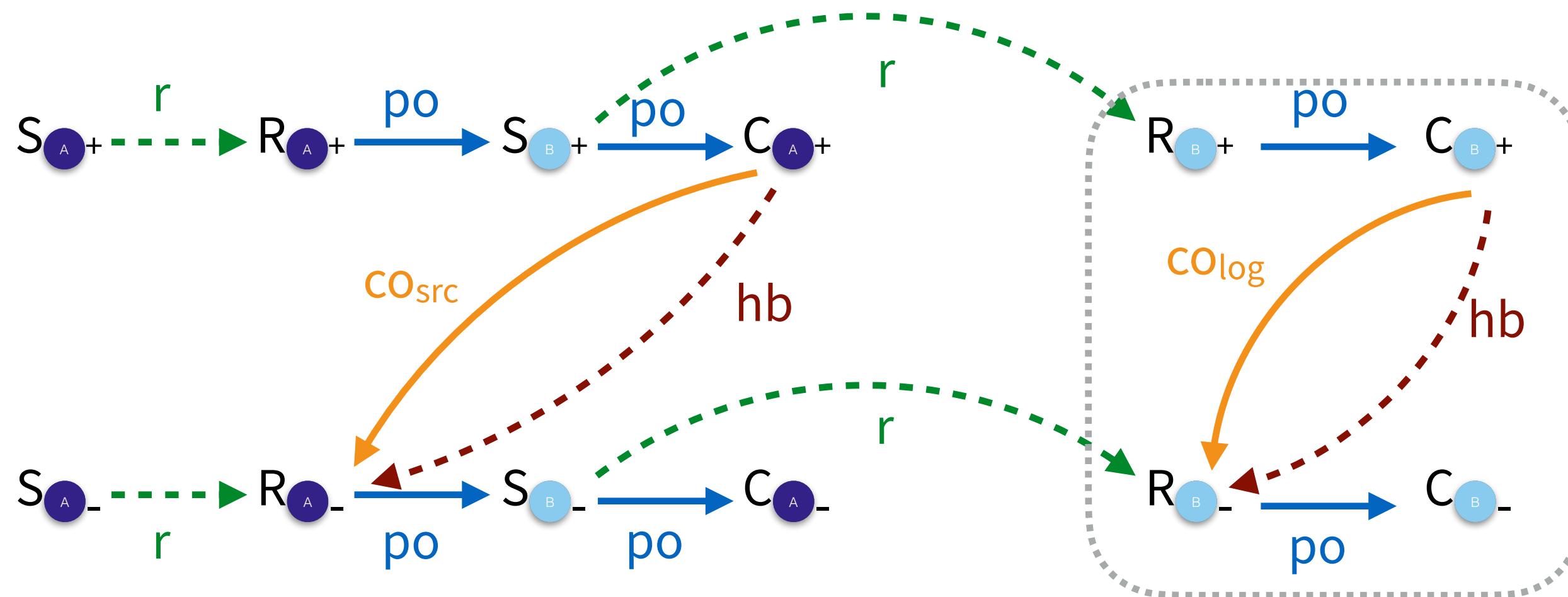


# Validating Causality

```
def update(account: cown[Account], amount: int) {
  when(account) { A
    if(account.balance + amount > 0) {
      account.balance += amount
      val id = account.id()
      when(log) { B log.record(id, amount) }
    }
  }
}
```

update(src, +100)

update(src, -100)





# What does this get us?

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